

Discriminative Prompt Rerouting and Query Prototype Refinement on Calibrated Prototype in Frozen Few Shot Class Incremental Learning

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Introduction: FSCIL Problem Formulation

N class of abundant images, K shots of new class images. Classification Problem

- Base session have many classes (>=100), many sample images per class
- Incremental session introduces new (novel) classes each having K(=5) labeled images
- Class sets across incremental sessions are mutually exclusive (called support sets)
- Old session data is inaccessible once training moves on
- Tested on unseen unlabeled query batch images

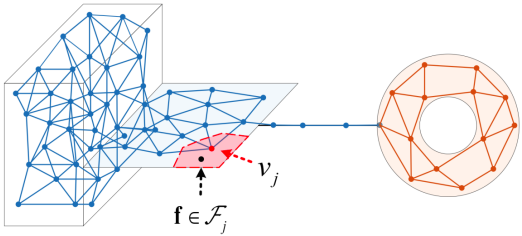


Figure-1: Base and Novel Class Prototypes in Feature Space visualization

Motivation: FSCIL Challenges

- **Catastrophic forgetting**: adapting to new classes overwrites representations learned for earlier classes, whereas focusing on possibly noisy support set forgets base prototypes.
- **Overfitting**: only 5 samples/class makes novel-class estimates unstable (Chen et al., CVPR'25).
- **Intransigence**: preserving knowledge from abundant existing classes from a highly limited dataset.
- **Stability-plasticity tradeoff**: continual fine-tuning chases plasticity (ability to learn new things) but forgets; incremental-session freezing buys stability (ability to preserve what already learnt) but sacrifices plasticity.
- **Risk of noisy prototypes**: No guard mechanism of noisy support set images.

Literature Review: CEC (Continually Evolved Classifiers) [2](CVPR, 2021)

- First, CEC trains a CNN backbone using the large base-class images and freeze the backbone
- Problem? - Classifiers from different sessions don't know about each other (Session 0 learns Cat, Dog forgets Tiger, Lion from session 1)
- Treat each class classifier weight as nodes in graph; learn relationship between all class nodes (Graph Attention network)
- Create preudo new classes (augmentation) from base classes to train the GAT; better generalization, better classification on novel class (Continual Pseudo Incremental Learning)

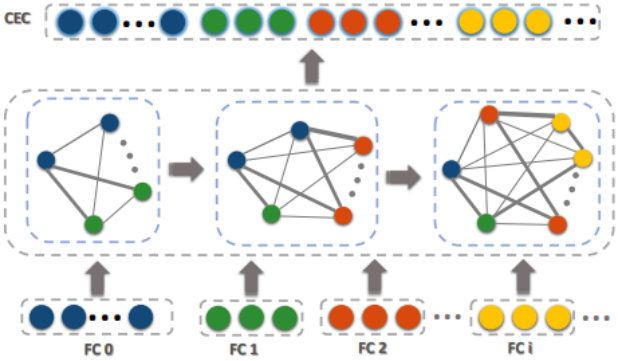


Figure-2: CEC graph model to adapt the classifier weights learned on individual sessions for the prediction over all classes

Literature Review: CEC (contd) [2](CVPR, 2021)

- Pretraining to learn the backbone model using training data in base session
- Pseudo incremental learning stage trains the graph model, require learnable parameters
- Classifier learning and adaptation stage uses support set (projection head weight update)

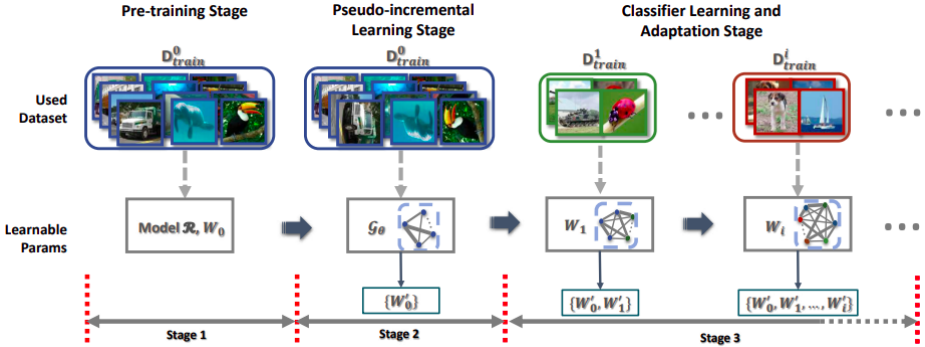


Figure-3: Pretraining the backbone for continual learning stages in CEC architecture

Literature Review: RKD (Relation Knowledge Distillation) [3](AAAI, 2021)

- Individual knowledge distillation (IKD) forces each old sample near its previous feature (overfit risk!)
- Instead of fixing each sample to a fixed place in feature space, RKD preserves the relational pattern across the class distribution.
- Think like instances can move, rotate, expand, or contract without destroying the relationship (angle); room for new classes

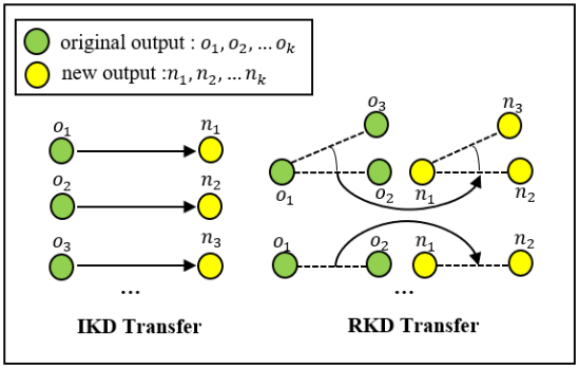


Figure-4: IKD transfers the knowledge point to point (original to new) while RKD transfers the structural relations (adjacency) of output

Building the Relation Graph require training at each incremental sessions

Literature Review: Semantic Knowledge Distillation [4](CVPR, 2021)

- Predicts a fixed-length **semantic word vector** for each image. A novel class only needs *its* word vector
- Base classes' word vectors are grouped together using k-means
- The idea is reusing knowledge from these semantically-related base-class word vectors
- An attention module learns how much to use each specialist embedding and combines them into the final representation
- Use these semantic knowledge via distillation to learn new classes

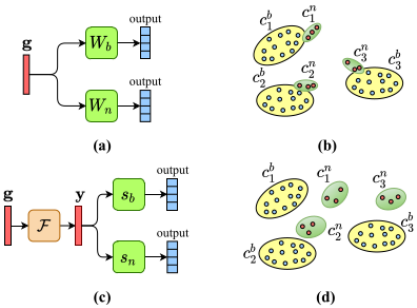


Figure-5: Semantically-guided network with vectors for well-separated base-novel classes



Figure-6: Novel class Hyena in presence of shared semantics (face, 4-ft height, body-shape, short-tail), helps classifying it and not to forget similar base classes (Dalmatian, Saluki, Meerkat)

Literature Review: FACT (ForwArd Compatible Training) [5](CVPR, 2022)

- In tradiotional knowledge-distillation, learn better base representation, then model learns to behave similarly at each incremental session (backward compatible)
- Instead of creating 5 prototypes from 5 base classes, create virtual class prototypes reserving spaces for future novel class (**forward compatible**)
- Intuition is that interpolations between clusters (mixture of classes: e.g., dog-cat mix) leads to low-confidence (possible future-class) regions (**prototype forcasting**)

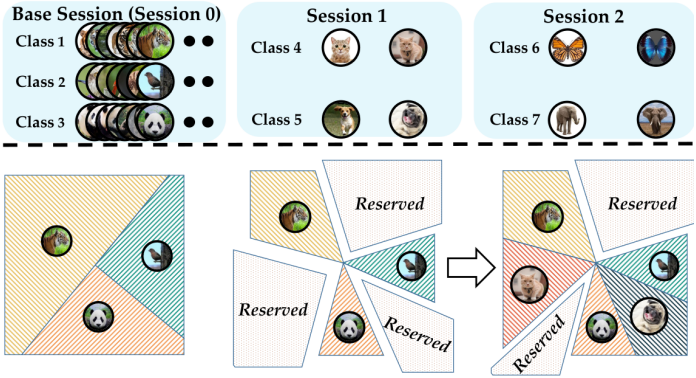


Figure-7: Traditional training vs reserved embedding space for novel classes in Forward Compatible training

Reserving spaces in feature space while creating virtual prototypes require substantial base classifier training using forward compatible loss

Literature Review: Semantic-Aware Virtual Contrastive Constraint [6](CVPR, 2023)

- "Better separation between base classes" leads to better ability to incorporate novel classes
- Classification and Contrastive loss are optimized in fantasy space (I don't have the future classes. But I can create additional semantic variations of the classes I do have)
- 12 augmentation (2 or 4-fold rotation x 3 color permutations) transformation on base classes; virtual class and fantasy label space for structuring huge empty spots in feature space

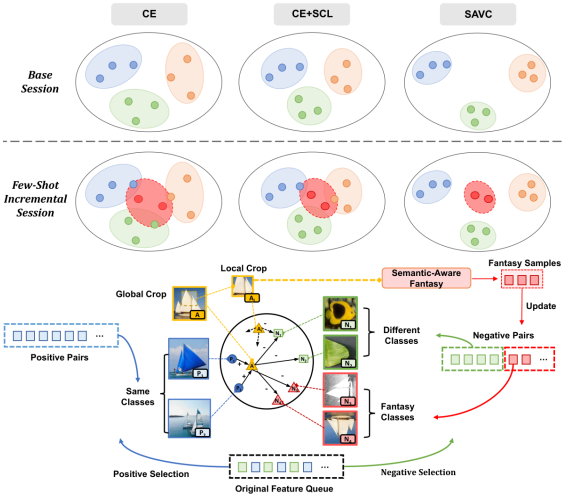
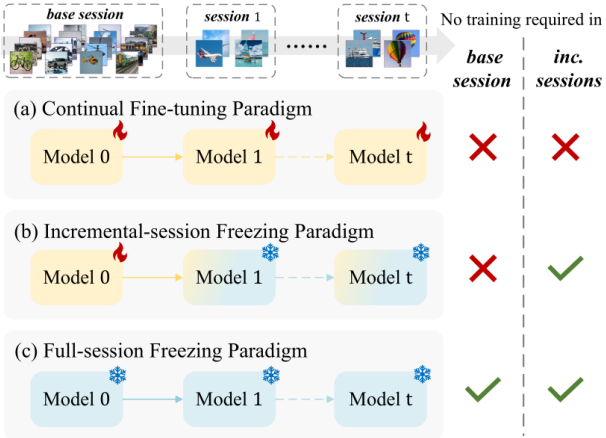


Figure-8: Embedding spaces comparison on different methods and semantic-aware virtual fantasy space

Towards Training Free FSCIL [7](CVPR, 2025)

- Prior methods solves plasticity-stability tradeoff via either both base and incremental session training or only base training and frozen support set
- Continual fine-tuning paradigm utilizes limited samples to continuously update the model, enhancing its plasticity.(**SKD, CEC, RKD,**)
- Incremental-session freezing focuses on learning representations during base task, generalized to incremental tasks (**SAVC, FACT, TEEN**)



Both require additional training to adapt to new tasks (resource efficiency? model adaptability?)

Literature Review: Bi-Level Modality Calibration (BiMC) [7](CVPR, 2025)

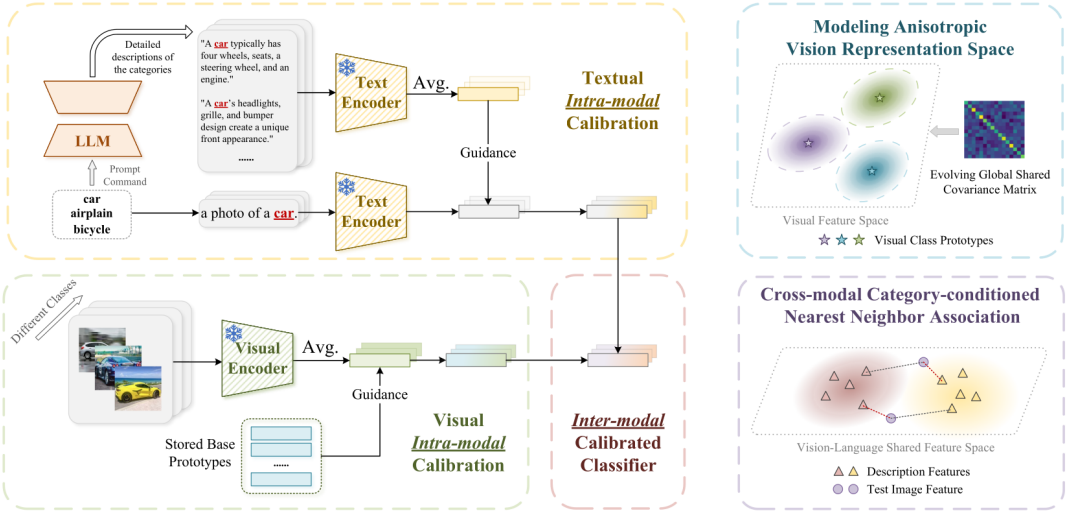


Figure-9: Bi-Level Modality Calibration Framework with globally shared covariance metric for visual feature modeling and category-conditioned nearest-neighbor scoring

Training-Free FSCIL: Everything so far...

Limitations of calibration in BiMC

- **Geometric** Only utilizing visual signals from CLIP is not suffiecient
- **Statistical** Novel prototypes are estimated only from few support images. No utilization of unlabeled query batches.

Our research questions

- Can we **increase the visual feature space** for more refined prototype and leave room for novel class prototypes without training?
- Can we rectify noisy novel-alike prototype after base session?
- Can we further increase the inter-class distance and reduce intra-class gap (more reserved space)?

Overview of Proposed Methodology

Frozen FSCIL is fundamentally a **three-level calibration problem**, added with prototype refinement strategies using qury batches and textual evidence.

⇒ **Inter-model calibration** with CLIP's share feature to produce stronger, complementary visual feature space (DINOv2) and **prototype-score calibration**

⇒ **Novel-mass gate** for novel prototype refinement utilizing **unlabeled query batch images**

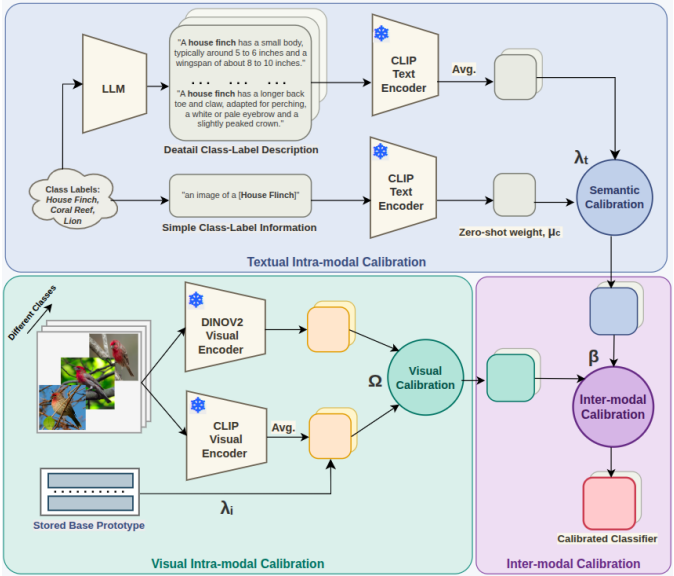
⇒ **Pairwise discriminative prompt matrix** to add a very little bias on calculated prototype

Methodology: Tri-Level Modality Calibration

1. Extract per class normalized visual features from both frozen encoder
2. Build semantic prototype using CLIP's text encoder
3. Inter-model visual calibration between CLIP and DINOv2: calculates each encoder's class-wise similarity scores, then combine scores
4. Finally, combine both semantic and visual evidence scores

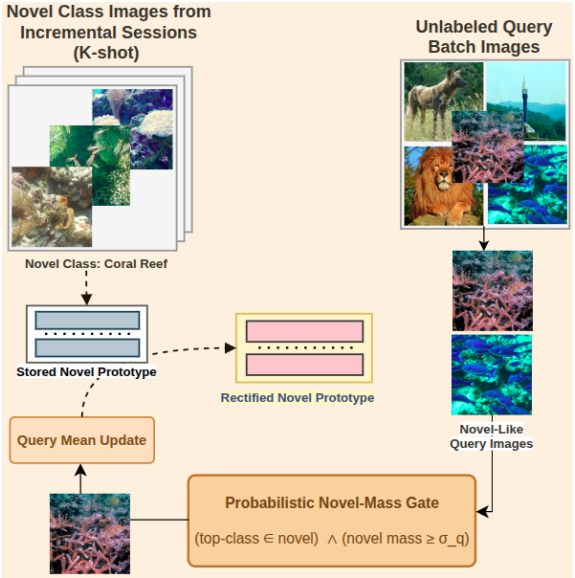
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[a] see Appendix for detailed formulation



Methodology: Query Prototype Refinement

1. Select reliable (noise-free) unlabeled query images and group them by predicted novel class
2. Compute a probability-weighted query prototype (more influence to higher-confidence queries)
3. Keep unreliable classes unchanged and blend reliable ones with noisy novel 5-shot prototypes



a

[a] see Appendix for detailed formulation

Methodology: Class Pairwise Discriminative Prompt matrix

For every class pair (i, j), matrix stores k=3 prompts describing class i *against* class j; each prompt targets a **different** discriminative attribute, the average text embedding $N_{i,j}$ spans the discriminative subspace rather than repeating one cue.

*mini*ImageNet $M_{\text{Saluki, Ibizan hound}}$

1. "a photo of a Saluki, whose long **feathered ears hang down** unlike the erect ears of an Ibizan hound"
2. "a photo of a Saluki, with **silky fringed fur** instead of the short hard coat of an Ibizan hound"
3. "a photo of a Saluki, showing a **dark nose and dark eyes** unlike a pink nosed Ibizan hound"

CUB-200 $M_{\text{Ring-billed Gull, Herring Gull}}$

1. "a photo of a Ring-billed Gull, notably **smaller and more delicate** than a Herring Gull"
2. "a photo of a Ring-billed Gull, with **yellow legs** rather than the pink legs of a Herring Gull"
3. "a photo of a Ring-billed Gull, its bill circled by a **black ring without any red spot**"



Dog-class: Saluki



Dog-class: Ibizan Hound



Bird-class: Herring Gull



Bird-class: Ring-billed Gull

Figure-10: Sample images from miniImageNet and CUB200

Methodology: Discriminative Prompt Rerouting (DPR)

Algorithm

1. Compute pairwise prototype similarity $S_{i,j}$ and new prototype vector of class i, $N_{i,j}$ following discriminative embeddings
2. Pick the most confusable unvisited pair (a, b) with $S_{a,b} > r_{th}$
3. Compute similarity of the *new* prototype to every other class i
4. Guard 1: keep only if $S_{a,b} > tmp_{a',b'}$
5. Guard 2: keep only if similarity to all other classes drops by < 0.1
6. Reroute prototype by updating T'_a, T'_b independently; until no pair exceeds r_{th}

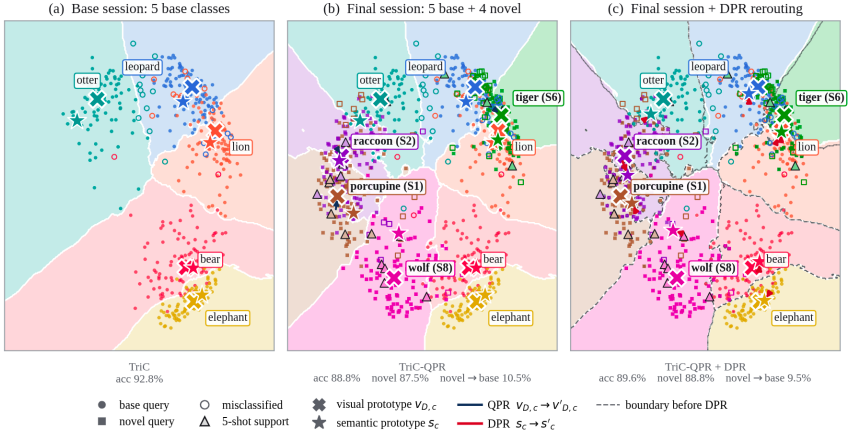


Figure-11: t-SNE heatmap of Prototype update with class-wise Discriminative Prompt

Results

[All values experimented with train-val-test split in **20 seeds** avg*]

Method	Modality	Accuracy in each session (%) ↑									Avg. ↑	PD ↓
		A_0	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8		
CLIP zero-shot	L	74.77	73.12	72.56	71.08	70.46	70.41	70.58	70.06	68.93	71.33	5.84
Visual prototype (CLIP)	V	75.88	72.78	71.01	68.52	66.55	64.84	64.12	62.71	61.03	67.49	14.85
TEEN	V	75.88	73.00	71.39	68.88	67.14	65.41	64.94	63.44	61.84	67.99	14.04
FeCAM	V	81.38	78.28	76.93	74.55	72.91	71.81	71.26	69.80	68.32	73.92	13.06
BiMC	V-L	79.70	77.97	77.29	75.35	74.76	74.39	74.34	73.55	72.50	75.54	7.20
BiMC Ens.	V-L	81.98	79.74	78.84	77.00	76.11	75.68	75.33	74.61	73.18	76.94	8.80
CLIP visual only	V	75.90	72.74	70.97	68.31	66.54	65.49	64.94	63.73	62.20	67.87	13.70
DINO visual only	V	89.63	87.87	87.21	85.16	85.22	84.90	84.92	84.57	83.34	85.87	6.29
OURS	V-L	90.90	89.78	89.42	88.04	88.17	87.98	88.12	87.91	87.13	**88.61**	**3.77**

Method	Modality	Accuracy in each session (%) ↑											Avg. ↑	PD ↓
		A_0	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	A_9	A_{10}		
CLIP zero-shot	L	66.38	64.05	62.71	59.07	59.24	59.27	57.76	56.69	55.46	55.19	55.73	59.23	10.65
Visual prototype (CLIP)	V	81.42	78.78	77.43	74.22	72.91	71.59	70.65	69.97	68.79	68.93	68.80	73.04	12.62
TEEN	V	81.42	79.16	77.49	74.60	73.26	71.75	70.68	70.21	68.90	69.28	69.12	73.26	12.30
FeCAM	V	82.26	79.48	77.76	74.68	72.86	71.10	69.92	69.17	67.61	67.89	67.40	72.74	14.86
BiMC	V-L	82.16	79.99	79.04	76.10	75.07	74.04	73.13	73.08	71.88	72.26	72.25	75.36	9.91
BiMC Ens.	V-L	83.00	81.01	79.83	76.96	75.84	74.73	73.86	73.47	72.26	72.61	72.68	76.02	10.32
CLIP visual only	V	81.63	79.14	77.49	74.24	73.26	71.68	70.81	70.03	68.53	68.72	68.47	73.09	13.16
DINO visual only	V	86.35	85.19	84.66	83.67	83.68	82.34	82.72	83.28	83.14	83.65	84.01	83.88	2.34
OURS	V-L	87.64	86.58	86.23	85.40	85.66	84.46	84.84	85.32	85.16	85.62	85.97	**85.72**	**1.67**

Table-1: Session-wise comparison on CIFAR100 and CUB200

Result Analysis: Tri level Calibration Impact

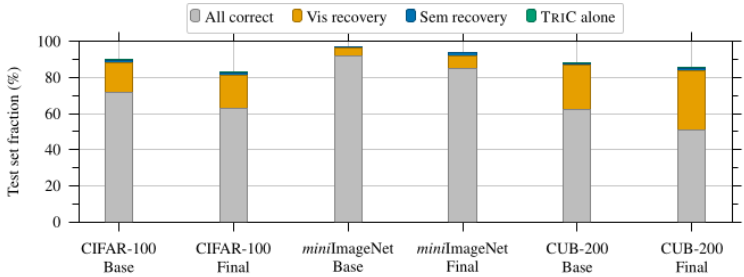


Figure-12: Per-query behavior of TRIC's inter-modal calibration at base and final sessions

Components			*mini*ImageNet accuracy in each session (%) ↑									Avg. ↑	PD ↓
Sem.	C-Vis.	D-Vis.	A_0	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8		
✓			93.22	93.18	91.79	91.40	91.18	90.69	89.32	89.40	89.26	91.05	3.96
	✓		92.58	91.86	89.61	88.82	88.75	88.16	86.93	86.82	86.79	88.92	5.79
		✓	96.42	96.02	94.47	94.38	94.43	93.49	92.30	92.07	91.90	93.94	4.52
✓	✓		94.90	94.80	93.21	92.91	92.88	92.60	91.78	91.88	91.81	92.97	3.09
✓		✓	97.06	96.95	95.67	95.66	95.70	95.19	94.37	94.24	94.09	95.44	2.97
	✓	✓	96.77	96.42	94.82	94.73	94.78	93.95	92.83	92.59	92.45	94.37	4.32
✓	✓	✓	97.25	97.16	95.85	95.83	95.87	95.42	94.62	94.47	94.34	95.65	2.91

Table-2: Component ablation of Calibration on miniImageNet with Semantic, CLIP-visual, and DINOv2-visual branches

Result Analysis: QPR Impact

Backbone	QPR	CIFAR-100			*mini*ImageNet			CUB-200		
		Full	Base	Novel	Full	Base	Novel	Full	Base	Novel
BiMC	off	72.61	75.21	68.70	91.88	93.15	89.98	72.27	80.08	64.63
BiMC	on	73.50	74.65	71.78	92.31	92.92	91.39	73.50	79.55	67.59
Δ		+0.89	−0.56	+3.08	+0.43	−0.23	+1.41	+1.23	−0.53	+2.96
BiMC + Ensemble	off	73.24	77.79	66.41	92.41	92.97	91.57	72.76	78.89	66.77
BiMC + Ensemble	on	73.28	76.73	68.10	92.55	92.68	92.37	73.01	77.70	68.42
Δ		+0.04	−1.06	+1.69	+0.14	−0.29	+0.80	+0.25	−1.19	+1.65
TRIC	off	83.07	88.48	74.96	93.73	96.13	90.12	85.29	87.84	82.80
TRIC	on	85.50	88.01	81.75	94.74	95.80	93.15	86.17	87.72	84.66
Δ		+2.43	−0.47	+6.79	+1.01	−0.33	+3.03	+0.88	−0.12	+1.86

Table-3: QPR Accuracy Analysis, Full/Base/Novel accuracy across different backbones

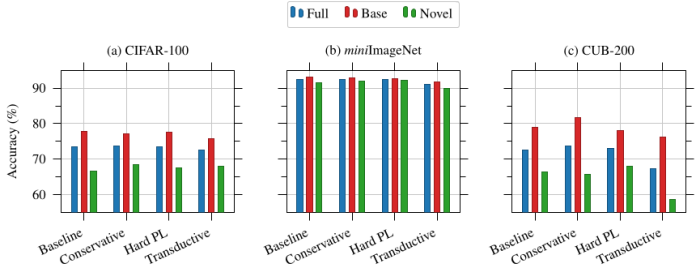


Figure-13: QPR ablation on different gating mechanisms (Baseline, Conservative, Hard PL, Transductive) over BiMC

Result Analysis: DPR Impact

Variant	Accuracy in each session (%) ↑									Avg. ↑	PD ↓
	A_0	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8		
BiMC	79.62	77.96	77.15	75.21	74.59	74.17	74.18	73.48	72.54	75.43	7.08
+ Ensemble	82.05	79.73	78.75	76.83	76.02	75.60	75.43	74.70	73.26	76.93	8.79
+ DINOv2 only	90.22	88.23	87.57	85.40	85.36	85.04	85.15	84.85	83.58	86.16	6.64
+ DINOv2/CLIP fusion	89.77	87.76	87.15	85.01	84.96	84.66	84.73	84.45	83.09	85.73	6.68
+ DPR	89.77	88.00	87.43	85.52	85.50	85.12	85.15	84.83	83.56	**86.10**	**6.21**
Δ from DPR	+0.00	+0.24	+0.28	+0.51	+0.54	+0.46	+0.42	+0.38	+0.47	+0.37	−0.47

Table-4: DPR ablation on per-session accuracy on each module with refined prototype in CIFAR100

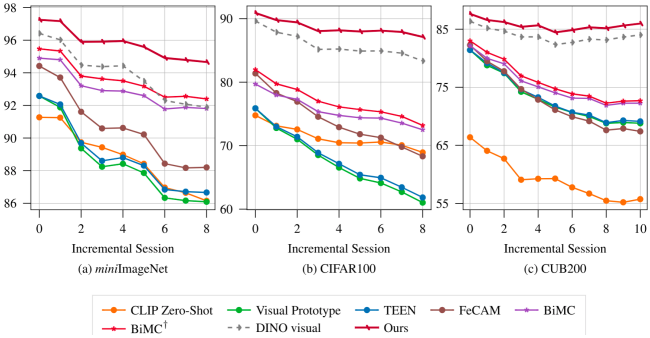


Figure-14: Final session-wise accuracy across all datasets: Ours vs Other methods [8](NeurIPS, 2023) [9](NeurIPS, 2024)

Conclusion

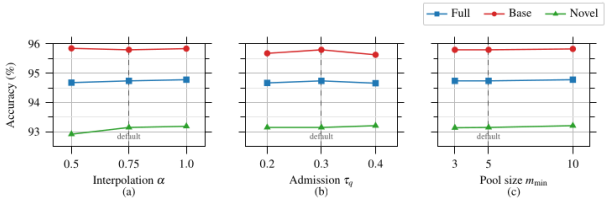
We propose a training-free approach to few-shot class-incremental learning where,

- Tri level calibration with stronger DinoV2 visual space is challenging as averaging incompatible embedding spaces is geometrically meaningless, so we calibrated [CLS] scores with normalized cosine distance
- QPR incorporates unlabeled query batch to rectify noisy novel prototypes, without leaking base-class noise which is well-motivated by the need for precision over coverage
- DPR reroutes only the most confusing class pair prototypes after calibration, thus creating virtual prototypes with class-specific discriminative-prompt. Update only when no other class's similarity gets worse because of rerouting

Limitations

Dataset	Gate purity	Base leakage	Gated queries	Updated novel classes
CIFAR-100	97.79%	2.21%	3191.7±27.8	40/40
*mini*ImageNet	98.57%	1.43%	3743.1±25.8	40/40
CUB-200	99.38%	0.62%	2861.2±8.0	99/100

(A) Prototype update statistics per query



(B) Hyperparameter sensitivity test on *mini*ImageNet, varied one at a time with default $(\alpha, \tau_q, m_{\min}) = (0.75, 0.3, 5)$

Limitations of our methods:

- QPR is transductive and may depend on query-batch composition.
- DPR depends on the quality of LLM generated discriminative prompts.
- Multiple coefficients are introduced, too much knob (not generalized). The hyperparameter selection is empirical.

Although, In our defence:

- QPR and DPR both module may not guarantee prototype update in all cases but on avg it performs better than baseline (20 seeds results show at least as good as baseline, on avg even better!)
- Hyperparameter sensitivity test shows that the effect of its change is negligible

References

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Appendix: Equations and Formulas in Methodology

TriC

$$\mathcal{C}_{\le t} = \bigcup_{s=0}^{t} \mathcal{C}^{(s)}, \quad \mathcal{C}_b = \mathcal{C}^{(0)}, \quad \mathcal{C}_n = \mathcal{C}_{\le t} \setminus \mathcal{C}_b$$

$$\mathrm{norm}(\mathbf{u}) = \frac{\mathbf{u}}{\|\mathbf{u}\|_2}, \quad q_C(\mathbf{x}) = \mathrm{norm}(f_I(\mathbf{x})), \quad q_D(\mathbf{x}) = \mathrm{norm}(f_D(\mathbf{x}))$$

$$w_c = \mathrm{norm}(f_T(\pi_c)), \quad \bar{z}_c = \mathrm{norm}\Big(\tfrac{1}{n_c} \textstyle\sum_{r=1}^{n_c} z_{c,r}\Big)$$

$$v_{C,c} = \mathrm{norm}\Big(\tfrac{1}{m_c} \textstyle\sum_j q_C(\mathbf{x}_{c,j})\Big), \quad v_{D,c} = \mathrm{norm}\Big(\tfrac{1}{m_c} \textstyle\sum_j q_D(\mathbf{x}_{c,j})\Big)$$

$$s_c = \mathrm{norm}\big((1-\lambda_T) w_c + \lambda_T \bar{z}_c\big)$$

$$\ell_c^S(\mathbf{x}) = q_C(\mathbf{x})^\top s_c$$

$$\ell_c^V(\mathbf{x}) = \omega\, q_D(\mathbf{x})^\top v_{D,c} + (1-\omega)\, q_C(\mathbf{x})^\top v_{C,c}$$

$$\ell_c^F(\mathbf{x}) = \beta \ell_c^S(\mathbf{x}) + (1-\beta)\ell_c^V(\mathbf{x}), \quad p_{\mathrm{TriC}}(c \mid \mathbf{x}) = \frac{\exp(\rho\, \ell_c^F(\mathbf{x}))}{\sum_{k \in \mathcal{C}_{\le t}} \exp(\rho\, \ell_k^F(\mathbf{x}))}$$

QPR

$$\hat{y}(\mathbf{x}) = \arg\max_{k \in \mathcal{C}_{\le t}} p^*(k \mid \mathbf{x}), \quad m_n(\mathbf{x}) = \sum_{k \in \mathcal{C}_n} p^*(k \mid \mathbf{x})$$

$$\mathcal{G}_c = \{\mathbf{x} \in \mathcal{U}_t : \hat{y}(\mathbf{x}) = c,\ m_n(\mathbf{x}) \ge \tau_q\}, \quad c \in \mathcal{C}_n$$

$$a_c(\mathbf{x}) = \frac{p^*(c \mid \mathbf{x})}{\sum_{\mathbf{x}' \in \mathcal{G}_c} p^*(c \mid \mathbf{x}')}, \quad \tilde{v}_{D,c} = \mathrm{norm}\Big(\textstyle\sum_{\mathbf{x} \in \mathcal{G}_c} a_c(\mathbf{x})\, q_D(\mathbf{x})\Big)$$

$$v'_{D,c} = \begin{cases} \mathrm{norm}\big((1-\alpha) v_{D,c} + \alpha \tilde{v}_{D,c}\big), & c \in \mathcal{C}_n,\ |\mathcal{G}_c| \ge m_{\min} \\ v_{D,c}, & \text{otherwise} \end{cases}$$

DPR

$$tmp_{a',i} = \cos(T'_a, T_i)$$

$$T'_a = T_a + \alpha D_{a \to b}, \quad T'_b = T_b + \alpha D_{b \to a}$$

$$\left(\forall_{\substack{i \ne a \\ i \ne b}} S_{a,i} - \forall_{\substack{i \ne a' \\ i \ne b'}} tmp_{a',i} \right) > -0.1$$